Indirect reciprocity with inertial reputation updating

Storyline 1, how reputations change (final draft)

Introduction

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Reputations are built in two steps. Observers judge each action against a social norm, and they revise their view of the actor in light of that judgement[7,8]. Theory has identified norms that make reputation-based cooperation evolutionarily stable[7,9], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[7,10]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[11,12], and most leading-eight norms no longer maintain stable cooperation[13–16]. Reputations must then remain both accurate and shared, distinguishing those who cooperate from those who do not while different observers largely agree about who is who[17,18].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,9,19]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[13,20,21]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[19,22,23]. The harm done by such disputed judgements depends on how often they occur and on how strongly each one changes a reputation. Remedies can therefore act at either step of reputation formation, on how actions are judged or on how judgements are turned into reputations, provided that genuine defectors still lose their standing.

Most remedies act on the first step. Public institutions, gossip, empathetic evaluation and the synchronization of private opinions reduce how often observers disagree[18,21,24–26], and generous or lenient assessment reduces how often a disputed act is condemned[27,28]. At the second step, the standard framework lets each new assessment replace an individual's image, so a single disputed act of punishment overturns a standing built over many interactions[7,13]. Quantitative scores, simple records, graded reputations and the aggregation of several observations soften this impact and can make cooperation more robust[22,29–33]. Most of them, however, treat reputational change as a single symmetric process, in which positive and negative evidence enter through the same score, threshold or observation window and affect treatment only once a threshold is crossed. Human evaluation departs from both features. Impressions of others are revised asymmetrically, although the direction of the asymmetry depends on context[34–37]. Institutions, too, grade their response and treat the two directions differently, escalating sanctions only with repeated violations[38,39] and restoring an offender's standing only after years without a further offence[40]. How the direction of reputational change and the treatment of incomplete evidence affect the punishment dilemma, and with it the evolution of cooperation, remains unclear.

Here, we develop a model of indirect reciprocity in which reputations have inertia, and in which this

inertia can differ between losing and regaining standing (Fig. 1). An established reputation changes only after a run of contrary assessments. Negative assessments move an individual in good standing through a sequence of warning states before the reputation turns bad, and positive assessments move an individual in bad standing through recovery states before it turns good. The two paths can differ in length, so that losing and regaining standing may require different amounts of evidence. Each intermediate state is interpreted as good or bad with a probability that sets how strongly incomplete evidence already affects treatment. The model thus treats two properties of reputational change as separate design choices, namely how much evidence is needed to change a reputation in each direction and how incomplete evidence is expressed in the meantime.

We find that the allocation of inertia between the two directions, rather than its total amount, determines whether private observers agree with one another and still distinguish cooperators from defectors. Although inertia acts only on how judgements are turned into reputations, it also reduces disagreement among observers, who share no information. The beneficial direction depends on the norm. Compact architectures, selected solely by these reputational criteria, make cooperation stable and easy to reach under each of four classical norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all. We further show that the tolerance that protects justified punishers can also shelter opportunists who defect only occasionally, a vulnerability that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Reputation-based cooperation therefore depends not only on which behaviours a society judges as good, but also on how readily it revises those judgements in each direction and on how it treats individuals before a judgement changes.

Discussion additions

Two paragraphs to add to the Discussion. The first sets out the two routes out of the punishment dilemma. The second carries the core of Storyline 2.

Our analysis suggests a general way to organize remedies for the punishment dilemma. The harm that disputed judgements do to justified punishers depends on how often such judgements occur and on how strongly each one changes a reputation. Institutions, gossip, synchronization and generous assessment act on the first factor[18,24,25,27], whereas scores, aggregation and inertia act on the second[22,29]. The same separation appears in our theory. Near a resident population, the reputational cost of withholding help is the product of a term from the first step, the assessment contrast, which measures how differently observers judge cooperation and defection, and a term from the second step, the slope of the reputation map, which measures how strongly a change in assessments moves a reputation (Eq. (3)). The two routes are therefore complementary. Weak agreement can be offset by robust revision, as in our model, and fragile revision does little harm when information is public. Leniency, however, has a price at either step. Generous assessment and tolerant revision both protect deliberate defectors as well as justified punishers[27], as our comparison of inertia, aggregation and scores shows (Fig. 5a). A reputation system should therefore separate whether a single negative assessment can overturn a reputation, which inertia suppresses, from what it costs at the margin, which partial consequences restore. Disputed evidence should be costly but not catastrophic.

Our results also bear on how agreement about reputations arises. Agreement is usually attributed to mechanisms that align information across observers, such as institutions, gossip and opinion synchronization[18,24,25], or to observers who draw on several observations at once[22]. In our model, observers share nothing and update on one observation at a time, yet they largely agree. Two independent observers who each regard an individual as good with probability r disagree with probability $2r(1-r)$, so agreement follows whenever private judgements become reliable. Under the punishment dilemma, reliability is not automatic, because the assessments an individual receives depend on what observers think

of its recipients. Inertia produces it when it is placed in the direction that separates the assessments of cooperators from those of defectors. Otherwise observers can agree on the wrong verdict, as under one-step Shunning, where they regard nearly everyone as bad (Fig. 2). Agreement through private revision complements information sharing rather than replacing it. It remains available where shared information is costly or untrusted, as in polarized societies[22,41], and it can be combined with institutions or gossip where they exist.

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